Case Study 1: The Outdated Leave Policy

(a) Why outdated + why confident

Outdated: LLM trained on old data; no access to updated HR documents.

Confident: LLMs generate fluent text regardless of accuracy; they don't distinguish between certain and uncertain knowledge.

(b) Four stages of RAG pipeline

1. Retrieval – Search and fetch current policy from HR database
2. Augmentation – Combine policy + employee's question
3. Generation – LLM answers based on retrieved current policy
4. Citation – Return answer with source reference

(c) Still wrong after fix - cause and detection

Cause: HR database contains outdated or conflicting policy versions.

Detection: Spot-check answers against source documents; audit for version control and timestamps.

(d) LLM alone vs RAG

Situation: General questions like "What is paternity leave?"

Justification: Doesn't need current policy updates; general knowledge is sufficient; RAG adds unnecessary overhead.

Case Study 2: The Case That Did Not Exist

(a) What is the name of this failure, and what is the technical reason it happens?

Name: Hallucination (or confabulation).

Technical reason: LLMs generate text by predicting statistically likely tokens. They cannot verify factual accuracy. When a source doesn't exist in training data, the model invents a plausible-sounding citation matching the statistical pattern of real citations, without any verification mechanism.

(b) The analyst said the chatbot "was not lying". Is that a fair description? Explain.

No, not fair. A lie requires intentional deception and knowledge of truth. The chatbot has no concept of truth or falsity—it only pattern-matches. Better description: it was hallucinating or confabulating.

(c) How would a RAG-based research assistant reduce this risk? Name two specific benefits.

1. **Source grounding** – RAG retrieves only from real documents, preventing invented citations. Every claim is tied to an actual source.
2. **Verification and traceability** – Users can directly verify claims by checking the retrieved source document, enabling accountability.

(d) RAG is not a complete solution. Recommend two process controls the firm should add.

1. **Citation verification protocol** – Analysts must spot-check citations before sending work to clients.
2. **Source documentation mandate** – Every claim must include a verifiable reference (author, publication, date, URL) that the team has physically confirmed exists.

Case Study 3: The Travel Agent That Booked the Wrong Flight

(a) Describe the first three cycles in Observe → Think → Act terms

Cycle 1:

- Observe: Query "cheapest flight to Dubai under $500 next Friday" + Calendar lookup
- Think: Search flights to Dubai for that date under $500 budget
- Act: Call flight search tool

Cycle 2:

- Observe: Results return flight with price "38,000" (rupees, no currency label)
- Think: 38,000 < 500? Agent interprets as under budget

- Act: Select this flight as cheapest option

Cycle 3:

- Observe: Flight selected; payment system ready
- Think: Approve payment for selected flight
- Act: Process payment

(b) Identify the failure point and explain why the error became expensive

Failure point: Agent did not normalize currency. Flight search returned "38,000" in Indian rupees without a currency label. Agent's budget logic couldn't distinguish USD from INR.

Why expensive: Agent approved 38,000 rupees (≈$450+ USD) as under the $500 USD budget. Payment was processed before error was caught, locking in a cost far above intended budget.

(c) A colleague proposes: "Let the agent run extra verification loops until it is sure." Is this enough? Explain.

No. More verification loops won't fix the root problem: the agent keeps making the same currency comparison error. Need structural fixes: explicit currency labels from tools, automatic currency conversion, or human approval gates for high-value transactions.

(d) List two safeguards you would add to the design

1. **Currency normalization** – Require all monetary values from tools to include explicit currency labels; convert all prices to standard currency (USD) before budget comparisons.
2. **High-value transaction hold** – Require human approval for any booking or payment over a set threshold (e.g., >$200) before agent executes payment.

Case Study 4: The IT Helpdesk Assistant

(a) Which parts are retrieval (MC8) and which are agentic (MC9)?

Retrieval (MC8): Knowledge-base search that finds and returns the "reset password" article. This is pure information retrieval without taking action.

Agentic (MC9): The password reset action—the tool that actually changes the password on the system. This involves observing, thinking, and executing a system change.

(b) Identify two separate failures in the pilot incident, one from each area

Retrieval failure: Knowledge-base article was outdated and contained incorrect procedure. Articles lack version control or freshness validation.

Agentic failure: Agent didn't verify user identity correctly. It matched on "similar name" instead of asking for employee ID, email, or security question before executing an irreversible action.

(c) Sketch two cycles of a safer loop for the password-reset request

Cycle 1:

- Observe: User requests "reset my password"
- Think: Cannot proceed without identity confirmation
- Act: Ask user for employee ID and registered email

Cycle 2:

- Observe: User provides ID and email; agent retrieves current reset procedure
- Think: Procedure says send reset link to verified email
- Act: Send reset link; ask user to confirm receipt before completing

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(d) Name two guardrails you would put around this assistant before going live

1. **Multi-factor identity verification** – Require at least two of: employee ID, registered email, security question. Never proceed on name matching alone.
2. **Knowledge-base freshness control** – Tag articles with version dates; flag articles older than 6 months for manual review; disable outdated articles in the agent.